

A Coordinate-Sweep Lemma Answers Question 1 in “Fixed Points of Averages of Resolvents”

Run `fa_banach_001`, lane 8

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Abstract

Bauschke, Wang, and Wylie ask whether the fixed-point set of their Gauss–Seidel-type sweep $T = J_m R_m \cdots J_1 R_1$ equals their product-space solution set S . We answer the question affirmatively. The only needed observation is that each map $J_k R_k$ changes only coordinate k . More generally, a sweep through maps that update one distinct coordinate at a time has no fixed points beyond the common fixed points of its component updates. Combining this elementary lemma with equation (64) of the source gives

$$S = \text{Fix } T = \text{Fix}(J_m R_m \cdots J_1 R_1).$$

The proof works for every positive integer m and requires no metric, linearity, continuity, or monotonicity assumptions on the coordinate updates.

1 Source question

The source is H. H. Bauschke, X. Wang, and C. J. S. Wylie, *Fixed Points of Averages of Resolvents: Geometry and Algorithms*, arXiv:1102.1478v1 [1]. On printed page 19 (PDF page 19), Remark 4.6 asks, for $m \geq 3$:

Q1: Concerning (65), is it actually true that

$$S = \text{Fix } T = \text{Fix}(J_m \circ R_m \circ \cdots \circ J_1 \circ R_1)?$$

The source passage is reproduced in Figure 1.

For $I = \{1, \dots, m\}$, the source works on the product space $\mathbf{X} = X^m$. Its equations (62)–(63) define maps R_k, J_k such that, whenever $i \neq k$,

$$(R_k x)_i = x_i, \quad (J_k x)_i = x_i.$$

Thus $U_k := J_k \circ R_k$ changes at most the k -th coordinate. The two other source identities needed below are

$$S = \bigcap_{k=1}^m \text{Fix } U_k \quad (\text{source equation (64)}) \tag{1}$$

and

$$T = U_m \circ \cdots \circ U_1 \quad (\text{source equation (66)}). \tag{2}$$

Let us now list some open problems.

Remark 4.6 (open problems) Suppose that $m \geq 3$. We do not know the answers to the following questions.

Q1: Concerning (65), is it actually true that

$$(69) \quad \mathbf{S} = \text{Fix } \mathbf{T} = \text{Fix}(\mathbf{J}_m \circ \mathbf{R}_m \circ \cdots \circ \mathbf{J}_1 \circ \mathbf{R}_1) ?$$

Q2: Can one give simple sufficient or necessary conditions for the convergence of the heuristic algorithm, i.e., the iteration of $\mathbf{T}$, when $\text{Fix } \mathbf{T} \neq \emptyset$?

Q3: Under the most general assumption (13), we observed convergence in numerical experiments of the new rigorous algorithm even though there is no underlying theory—see Proposition 3.4(ii) and Theorem 4.1. Can one provide simple sufficient or necessary conditions for the convergence of the sequence defined by (59)?

Remark 4.7 Concerning Remark 4.6, we note that the first two questions posed have affirmative answers when $m = 2$. Indeed, one then computes

$$(70) \quad \mathbf{T}: \mathbf{X} \rightarrow \mathbf{X}: (x_1, x_2) \mapsto (J_{\lambda_2^{-1}A_1}x_2, J_{\lambda_1^{-1}A_2}J_{\lambda_2^{-1}A_1}x_2)$$

and hence $(x_1, x_2) \in \text{Fix } \mathbf{T}$ if and only if $x_1 = J_{\lambda_2^{-1}A_1}x_2$ and $x_2 = J_{\lambda_1^{-1}A_2}x_1$, which is the same as requiring that $(x_1, x_2) \in \mathbf{S}$. When $\mathbf{S} = \text{Fix } \mathbf{T} \neq \emptyset$, then the iterates of $\mathbf{T}$ converge weakly to a fixed point by [30, Theorem 5.3(i)].

Appendix A

Most of this part of the appendix is part of the folklore; however, we include it here for completeness and because we have not quite found a reference that makes all points we

Figure 1: Remark 4.6 and its context in arXiv:1102.1478v1, p. 19.

2 Proof intuition

Run one complete sweep, beginning with coordinate 1. Once coordinate 1 has been updated, no later map can alter it. Therefore, if the end of the sweep equals the starting vector, the first update must already have left the vector fixed. After removing that vacuous first update, the same argument applies to coordinate 2, and successively to all remaining coordinates. This triangular causality is the entire proof.

3 The coordinate-sweep theorem

Theorem 1 (Fixed points of a one-pass coordinate sweep). *Let $Y = Y_1 \times \cdots \times Y_m$ be a Cartesian product of arbitrary sets. For each $k \in \{1, \dots, m\}$, let $U_k: Y \rightarrow Y$ satisfy*

$$(U_k y)_i = y_i \quad (i \neq k). \quad (3)$$

Then

$$\text{Fix}(U_m \circ \cdots \circ U_1) = \bigcap_{k=1}^m \text{Fix } U_k. \quad (4)$$

Proof. The inclusion from right to left is immediate. For the reverse inclusion, take $y \in \text{Fix}(U_m \circ$

$\dots \circ U_1$), and define the intermediate states

$$y^{(0)} = y, \quad y^{(k)} = U_k y^{(k-1)} \quad (1 \leq k \leq m).$$

Thus $y^{(m)} = y$. We prove inductively that $y^{(k)} = y$ for every k .

The assertion holds for $k = 0$. Suppose $y^{(k-1)} = y$. By (3), applying U_k leaves every coordinate other than k unchanged, so

$$(y^{(k)})_i = y_i \quad (i \neq k).$$

Every later map $U_{k+1}, \dots, U_m$ also leaves coordinate k unchanged. Hence

$$(y^{(k)})_k = (y^{(m)})_k = y_k.$$

All coordinates of $y^{(k)}$ therefore equal those of y , proving $y^{(k)} = y$. In particular, $U_k y = U_k y^{(k-1)} = y^{(k)} = y$. The induction gives $y \in \bigcap_k \text{Fix } U_k$, as required. $\square$

Corollary 2 (Affirmative answer to Question 1). *In the setting of arXiv:1102.1478,*

$$S = \text{Fix } T = \text{Fix}(J_m \circ R_m \circ \dots \circ J_1 \circ R_1).$$

In particular, Question 1 in Remark 4.6 has an affirmative answer.

Proof. Set $U_k = J_k \circ R_k$. Source equations (62)–(63) give (3), so Theorem 1 and (2) yield

$$\text{Fix } T = \bigcap_{k=1}^m \text{Fix}(J_k \circ R_k).$$

The right-hand side is S by (1). $\square$

4 Verification, scope, and novelty

Definition and order audit. The proof uses the source’s actual order $T = U_m \circ \dots \circ U_1$: the first applied map is U_1 , and every later map leaves coordinate 1 unchanged. Equations (62) and (63) separately leave all non- k coordinates unchanged, so their composition does as well. Equation (64) is exactly the common-fixed-point identity needed in the last line. The argument also remains valid when these sets are empty.

Strength and limitation. The abstract theorem applies to every $m \geq 1$, to any order that visits each coordinate exactly once, and to maps between products of arbitrary sets. These strengthenings are incidental. The result answers only Q1 of Remark 4.6. It supplies no convergence theorem for iterates of T and does not answer Q2 or Q3.

Novelty check. Cheap run and corpus indexes, followed by bounded searches using the arXiv identifier, paper title, displayed equality, and core fixed-point and Gauss–Seidel terminology, found no later general answer. A later paper, *On moving averages* [2], rigorously analyzes the algorithm in a special case; its abstract describes the result as a special-case convergence proof. This search is evidence, not an exhaustive bibliographic guarantee. The result should therefore remain a candidate full solution pending independent mathematical and literature review.

References

- [1] H. H. Bauschke, X. Wang, and C. J. S. Wylie, *Fixed Points of Averages of Resolvents: Geometry and Algorithms*, arXiv:1102.1478v1 (2011), <https://arxiv.org/abs/1102.1478>.
- [2] H. H. Bauschke, J. Sarada, and X. Wang, *On moving averages*, arXiv:1206.3610 (2012), <https://arxiv.org/abs/1206.3610>.